



Customer Shopping Behavior Analysis

Uncovering spending patterns, segments, and product preferences from **3,900 transactions** to drive strategic decisions.

Dataset at a Glance

3,900

Transactions

18

Columns

37

Missing Values

Review Rating only

25

Product Types

Features span demographics, purchase details, and shopping behavior across 4 categories.

Data Preparation in Python

Clean

Imputed missing Review Ratings by category median

Standardize

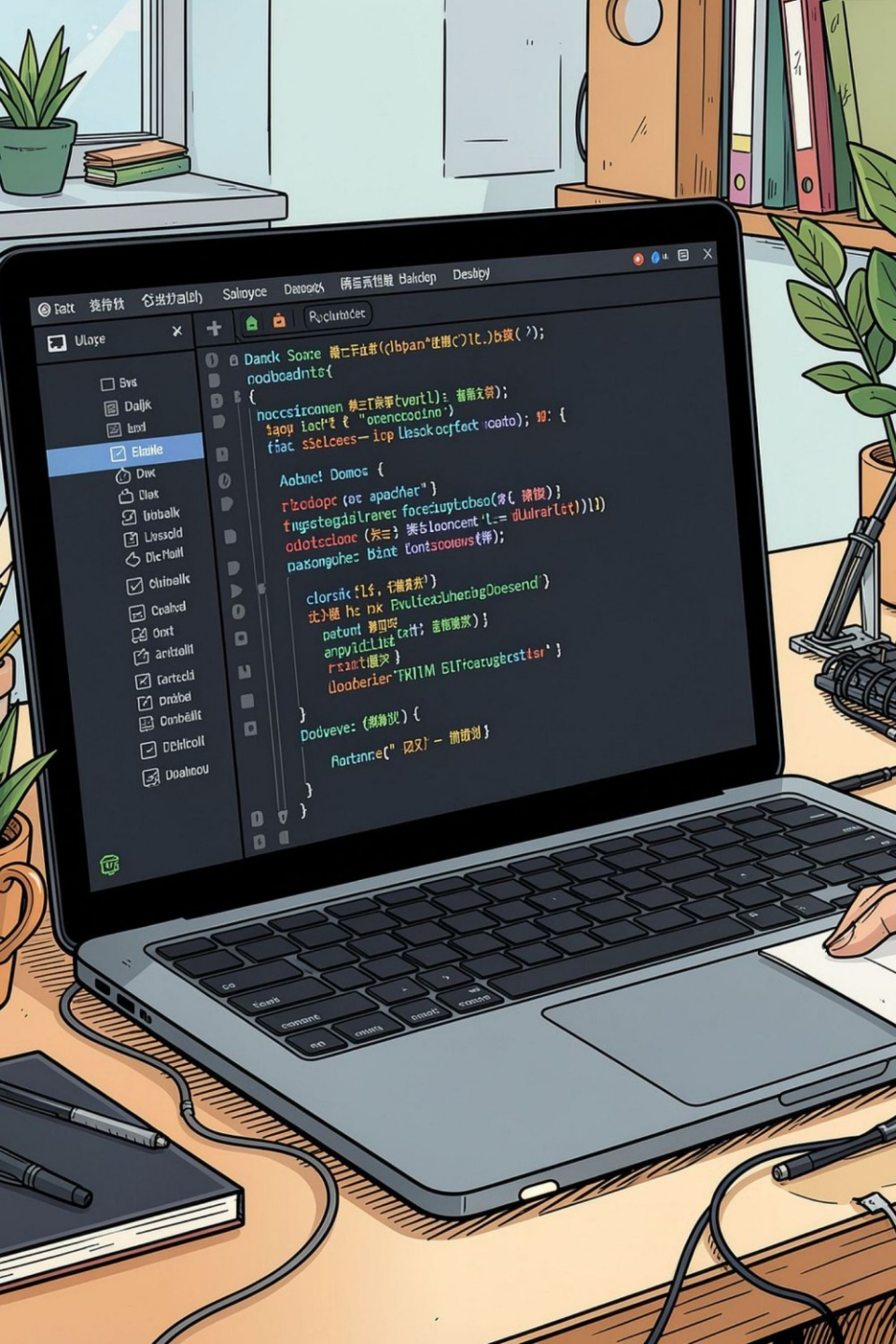
Renamed columns to snake_case; dropped redundant fields

Engineer

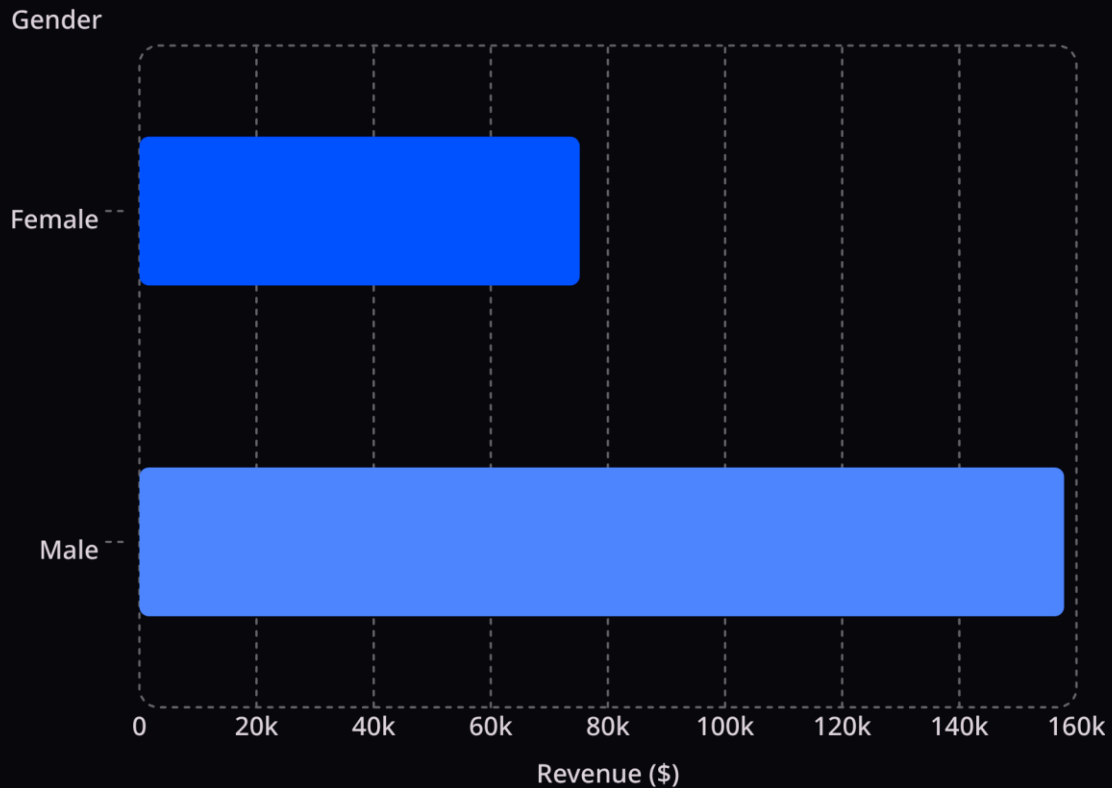
Created age_group and purchase_frequency_days features

Load

Connected to SQL Server for structured analysis



Revenue by Gender

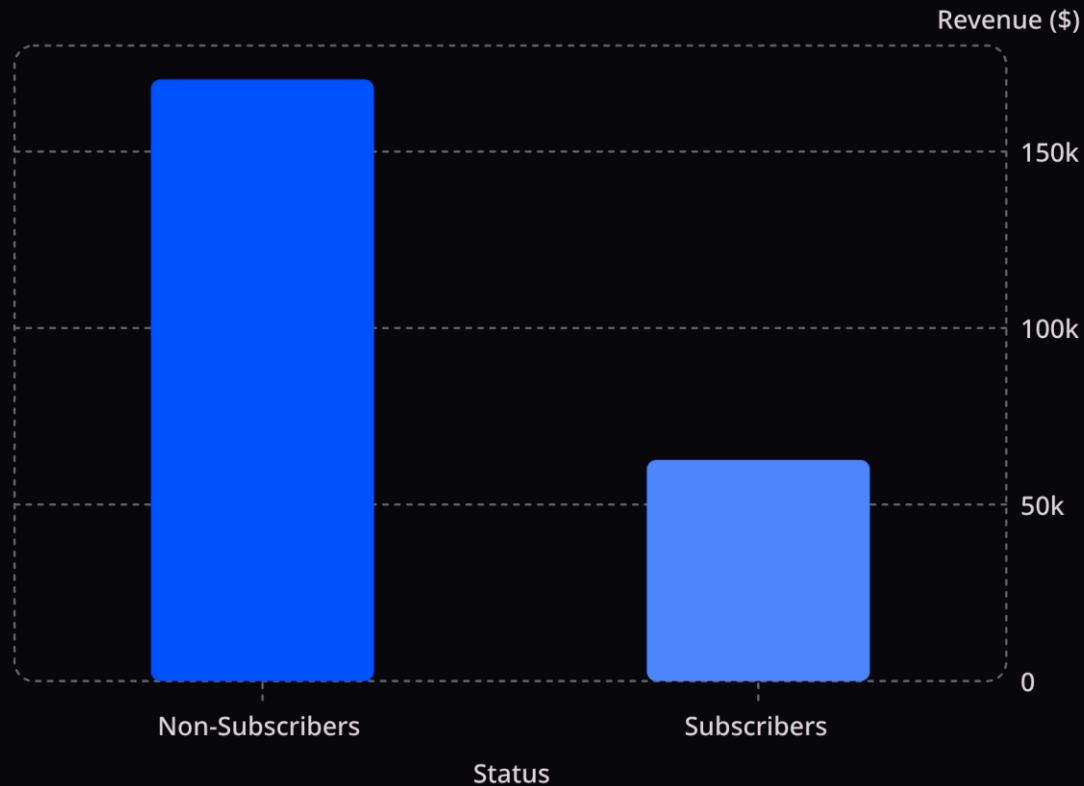


Key Finding

Male customers generate **2.1× more revenue** than female customers.

- Male: \$157,890 total
- Female: \$75,191 total

Subscribers vs. Non-Subscribers

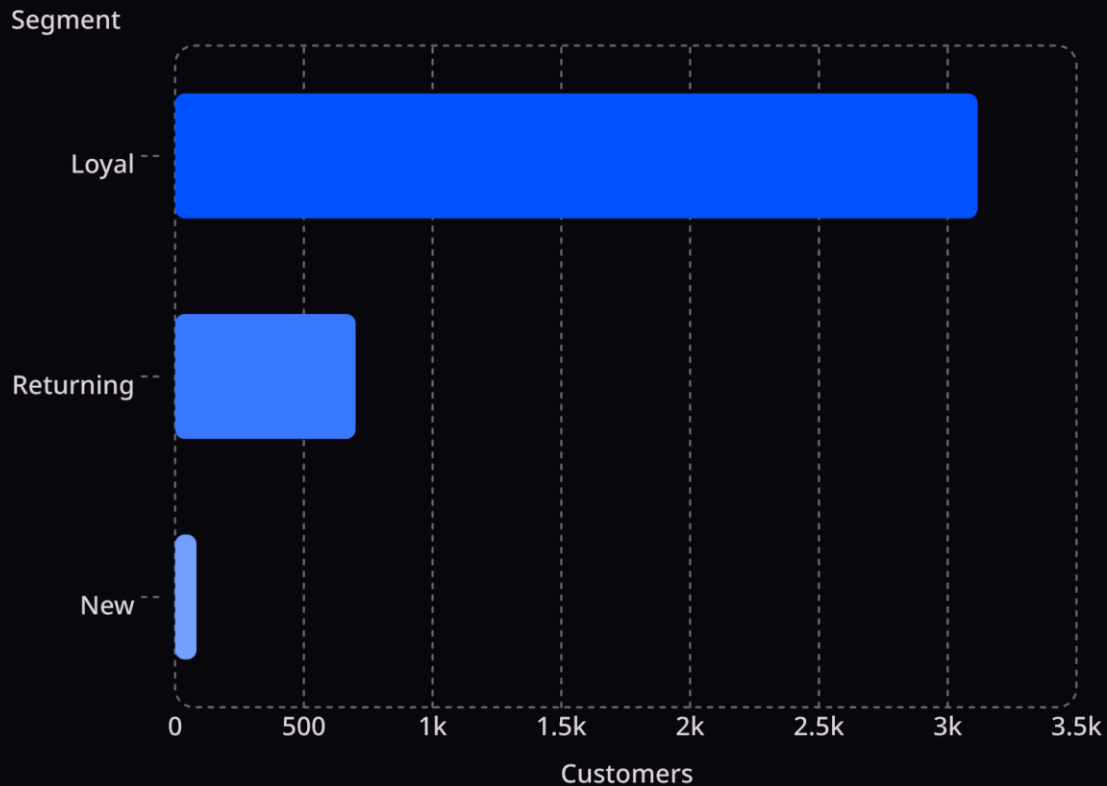


Subscription Breakdown

- **73%** non-subscribers — \$170K revenue
- **27%** subscribers — \$63K revenue
- Avg spend nearly identical (~\$59)

📌 958 repeat buyers are subscribed — strong loyalty signal.

Customer Segmentation

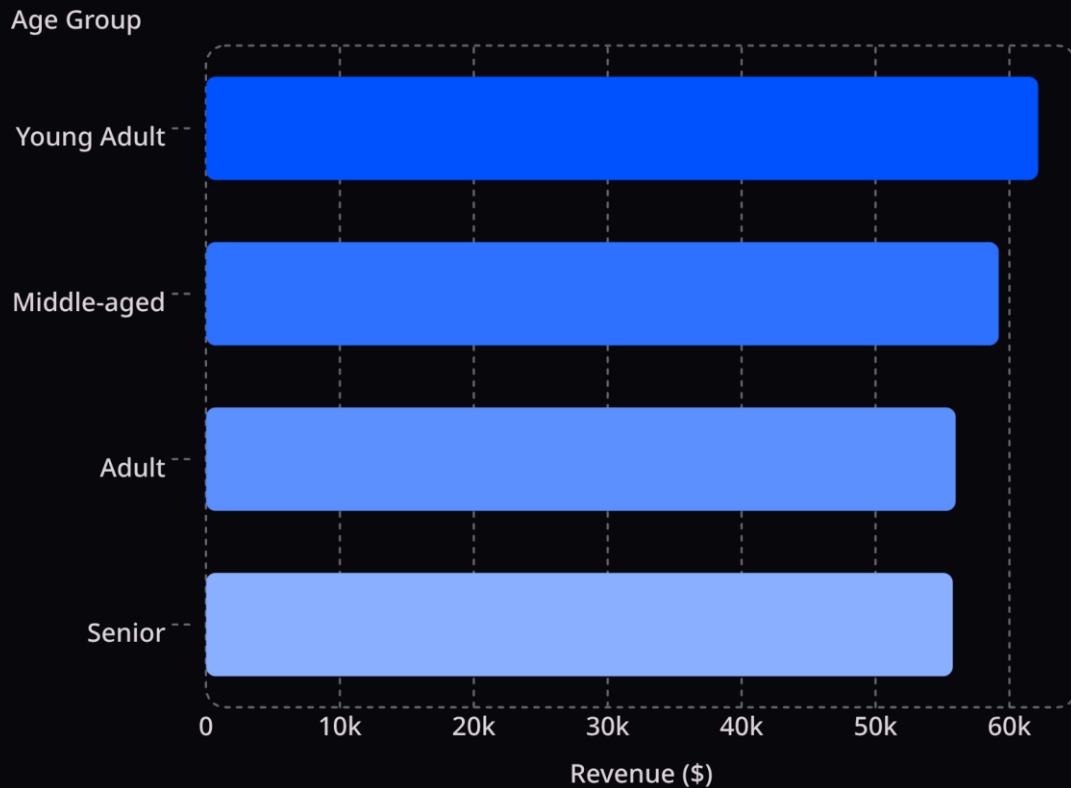


Segment Distribution

- **Loyal:** 3,116 customers (80%)
- **Returning:** 701 customers (18%)
- **New:** 83 customers (2%)

Strong loyalty base — opportunity to convert Returning → Loyal.

Revenue by Age Group



Age Insights

Young Adults lead revenue at **\$62,143**, but all groups contribute relatively evenly.

- Young Adult: \$62K
- Middle-aged: \$59K
- Adult: \$56K
- Senior: \$56K

Top Products & Discount Patterns

Top Rated Products

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Highest Discount Rates

Product	Discount %
Hat	50%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%



Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27%

Subscribers

Interactive filters: Subscription Status, Gender, Category, Shipping Type.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers

